

Topic C6

Chemical reactions

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

C6.1	Physical and chemical changes
C6.2	Rate of reaction
C6.3	Reversible reactions and equilibrium
C6.4	Redox

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Physical and chemical changes

A physical change does not form a new substance and is often reversible by physical means. A chemical change rearranges atoms and forms new substances; signs can include gas, precipitate, lasting colour change or temperature change, but evidence should be interpreted in context. Mass is conserved in a closed system.

Examiner insight

Melting is physical even though appearance changes; burning is chemical because new substances form.

2. Collision theory and rate

Particles react only when they collide with sufficient energy and suitable orientation. Rate increases with concentration or gas pressure because collisions are more frequent; with surface area because more particles are exposed; and with temperature because collisions are more frequent and a larger fraction exceed activation energy.

rate = quantity of reactant used or product formed / time

Examiner insight

For temperature, mention both faster collisions and a greater proportion above activation energy.

3. Catalysts and rate experiments

A catalyst increases rate by providing a lower-activation-energy pathway and remains chemically unchanged at the end. Rates can be followed by gas volume, mass loss, colour change or precipitate formation. Initial rate is the tangent gradient at time zero; a plateau shows reaction completion.

Worked example

If 48 cm³ gas forms in 60 s, mean rate = $48/60 = 0.80$ cm³/s.

Examiner insight

A catalyst changes how fast equilibrium is reached but not equilibrium composition.

4. Reversible reactions and equilibrium

A reversible reaction can proceed in both directions. Dynamic equilibrium occurs in a closed system when forward and reverse rates are equal; concentrations remain constant, not necessarily equal. Increasing temperature favours the endothermic direction. Increasing pressure favours the side with fewer gas molecules. Removing a product shifts equilibrium to replace it.

Examiner insight

Always identify the change, the direction that opposes it, and the resulting yield.

5. Redox

Oxidation is gain of oxygen or loss of electrons; reduction is loss of oxygen or gain of electrons. The reducing agent is oxidised and donates electrons; the oxidising agent is reduced and accepts electrons. Oxidation states provide another way to track electron transfer.

Worked example

$\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$: CuO is reduced; H_2 is oxidised and is the reducing agent.

Examiner insight

Name both the species and the change; avoid saying only “it is redox”.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

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hemical reactions

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which is a chemical change? [1]
- A melting ice
 - B boiling water
 - C burning magnesium
 - D dissolving sugar
2. Rate can be calculated as [1]
- A time/change
 - B change/time
 - C mass×time
 - D volume+time
3. Why does higher concentration increase rate? [1]
- A Particles get larger.
 - B More frequent collisions occur.
 - C Activation energy increases.
 - D Bonds vanish.
4. Why does higher temperature have a large effect? [1]
- A Fewer particles move.
 - B More particles exceed activation energy.
 - C Concentration becomes zero.
 - D Catalyst is formed.
5. A catalyst [1]
- A increases Delta H
 - B is consumed completely
 - C lowers activation energy
 - D increases product energy
6. Dynamic equilibrium requires a [1]
- A closed system
 - B solid only
 - C zero forward rate
 - D catalyst always
7. At equilibrium, forward and reverse reactions have [1]
- A equal concentrations
 - B equal rates
 - C zero rates
 - D equal activation energies

8. Increasing pressure favours the side with [1]
- A more gas molecules
 - B fewer gas molecules
 - C more solids
 - D higher temperature
9. Oxidation is [1]
- A gain of electrons
 - B loss of electrons
 - C loss of oxygen only
 - D decrease in oxidation state
10. An oxidising agent [1]
- A is oxidised
 - B donates electrons
 - C accepts electrons and is reduced
 - D must contain oxygen

Section B · Structured questions

11 Marble chips react with dilute hydrochloric acid and produce carbon dioxide.

(a) Write the word equation.

[2]

(b) State two ways to measure rate.

[2]

(c) Explain why powdered marble reacts faster.

[2]

(d) State why final gas volume is unchanged if only particle size changes.

[2]

[Total: 8]

12 A student obtains a gas-volume against time graph.

(a) Describe how rate changes during the reaction.

[2]

(b) Explain the change using concentration.

[2]

(c) Explain what the plateau means.

[2]

(d) Describe how to determine initial rate. [2]

[Total: 8]

13 The Haber reaction is $\text{N}_2 + 3\text{H}_2$ reversible to 2NH_3 and is exothermic forward.

(a) Define dynamic equilibrium. [3]

(b) Predict effect of increasing pressure on ammonia yield. [2]

(c) Predict effect of increasing temperature on yield. [2]

(d) State the effect of a catalyst on equilibrium yield. [1]

[Total: 8]

14 Copper(II) oxide reacts with carbon: $2\text{CuO} + \text{C} \rightarrow 2\text{Cu} + \text{CO}_2$.

(a) State what is oxidised. [1]

(b) State what is reduced. [1]

(c) Identify the reducing agent and explain. [3]

(d) Explain why the reaction is redox.

[3]

[Total: 8]

15 A catalyst is added to an exothermic reversible reaction at equilibrium.

(a) Explain how it increases forward rate.

[2]

(b) Explain its effect on reverse rate.

[2]

(c) State the effect on equilibrium composition.

[2]

(d) Explain why equilibrium is reached sooner.

[2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 C. Magnesium oxide, a new substance, forms.
- 2 B. Rate is change in quantity per unit time.
- 3 B. More particles per volume collide more often.
- 4 B. The energetic fraction rises.
- 5 C. It provides an alternative pathway.
- 6 A. Matter must not enter or leave.
- 7 B. Rates are equal while reactions continue.
- 8 B. The shift reduces pressure.
- 9 B. Electron loss is oxidation.
- 10 C. It causes oxidation by accepting electrons.

Solutions · Structured questions

11. 11(a) calcium carbonate + hydrochloric acid $\rightarrow$ calcium chloride + water + carbon dioxide [2]. (b) Gas volume/time or mass loss/time [2]. (c) Greater surface area gives more frequent successful collisions [2]. (d) Same reactant moles and limiting reagent, so same product amount [2].
12. 12(a) Fast initially then slows to zero [2]. (b) Reactants are used up, concentration and collision frequency fall [2]. (c) Limiting reactant is exhausted / no more gas forms [2]. (d) Draw tangent at time zero and calculate gradient [2].
13. 13(a) Closed system; forward and reverse reactions continue at equal rates; concentrations constant [3]. (b) Yield increases because product side has fewer gas molecules [2]. (c) Yield decreases because equilibrium favours endothermic reverse direction [2]. (d) No change [1].
14. 14(a) Carbon [1]. (b) Copper(II) oxide [1]. (c) Carbon; it removes oxygen from CuO and gains oxygen [3]. (d) Oxidation and reduction occur together / oxygen transfers from CuO to C [3].
15. 15(a) Alternative pathway with lower activation energy [2]. (b) Reverse activation energy is also lowered and reverse rate increases [2]. (c) No change [2]. (d) Both reactions proceed faster until equal rates are re-established [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.